

QGIS Workflow

1. Multi-Band Stacking

This step enriches the raw imagery by combining standard optical bands with NDVI, which is helpful for distinguishing red-painted asphalt from bare soil or dead vegetation.

- **Tool: Concatenate images**
- **Inputs:** raw 4-band orthophoto (Red, Green, Blue, NIR) and 1-band NDVI raster previously calculated from it.
- **Result:** A unified 5-band raster (R, G, B, NIR, NDVI).

2. Machine Learning

To prevent IndexError crashes, the classifier is trained and executed on the unclipped 5-band raster. Using the Imbalanced Guard ensures that the underrepresented bikelane pixels are not statistically overwhelmed by massive background classes.

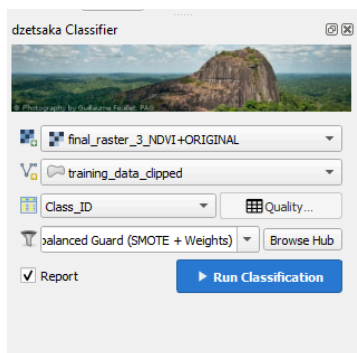
2.1. Manual Training Data Digitization

Manually created a new vector layer and digitized pure sample polygons representing the core materials. Integer IDs were assigned to several distinct target classes: **wet painted bikelanes (separated)**, **paved bikelanes (unseparated)**, **asphalt**, and **background**.

2.2. Model Training and Prediction

Tool: dzetsaka Classifier

- Raster to classify: unmasked 5-band raster.
- Training vector layer: manually digitized vector training layer.
- Class label field: Class_ID (integer labels for each material).
- Classification recipe: RF Imbalanced Guard (SMOTE + Weights)
- Before launching the process, dzetsaka Classifier allows to check training data quality for common issues: class imbalance, insufficient samples, spatial clustering.



dzetsaka Classifier interface

3. Area Filtering

This stage defines the geographic limits of the analysis, restricting the output to the actual street network.

3.1. Network Acquisition

- Tool: QuickOSM

3.2. Buffer Generation

- Tool: Buffer
- Inputs: linear OSM street network.
- Key Parameters:
 - Distance: e.g., 4.5 m for cycleways or 6 m for standard lanes.
 - Dissolve result: True.

Result: a mask of region of interest.

4. Post-Classification Masking

We now cleanly cut the raw classification raster using the refined spatial mask, eliminating all physical obstacles and off-road predictions.

- **Tool:** Clip Raster by Mask Layer
- **Inputs:**
 - Input layer: raw Random Forest classification raster (from Stage 2).
 - Mask layer: mask of region of interest (from Stage 3).
- **Key Parameters:**
 - Assign a specified nodata value to output bands: **0**.
- **Result:** A clean classification raster restricted strictly to valid, unshaded pavement within the road network.

5. Object-Based Segmentation and Class Assignment

The final phase eliminates pixelated noise by grouping pixels into structurally meaningful geometric shapes, and assigning the dominant ML class to each object.

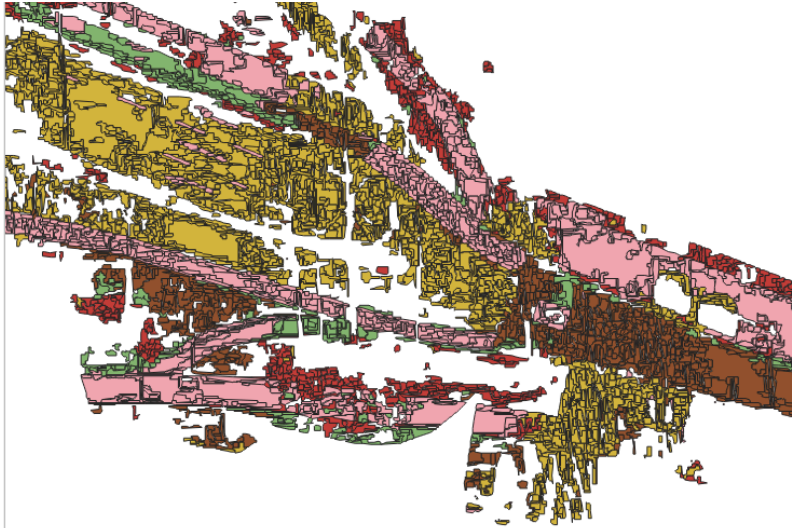
5.1. Vector Segmentation

- **Tool:** OTB Segmentation (meanshift)
- **Input:** the 5-band raster.
- **Algorithm:** Meanshift
- Spatial radius: **5** (controls the spatial search range).
- Range radius: **15**.
- Minimum region size: **50** (absorbs tiny, noisy segments).
- **Result:** a vector layer of thousands of homogeneous polygons outlining road markings, asphalt patches, shadows etc.

5.2: Class Assignment

- **Tool:** Zonal Statistics
- **Inputs:** MeanShift vector polygons (from 5.1) and masked classification raster (from Stage 4).
- **Key Parameters:**
 - Statistics to calculate: **Majority** (only).
 - Column prefix: class_
- **Result:** The final dataset. Each structural vector polygon is assigned the material class

that covers the majority of its internal area, resulting in features ready for cartographic visualization.



Example map of final dataset